

ElectroDSL

All built-in and building electrical symbols with exact DSL syntax, pin names, wiring rules and prompt templates

File extension	.edsl – plain text, open with any editor or ElectroDSL
Circuit block	CIRCUIT "Title" { ... }
Component line	ID: Type(params) at (x, y) label="Name" [value="val"] [rotate=0 90 180 270]
Wire line	wire FROM_ID.pinName -> TO_ID.pinName
Chained wire	connect A.pin -> B.pin -> C.pin -> D.pin
Grid coords	x: 0-14 (left→right) y: 0-10 (bottom→top)
AI output rule	ONLY raw DSL starting with CIRCUIT – no markdown, no backticks

1. Complete DSL Syntax Reference

1.1 File Structure

[illegible]

1.2 Component Line Fields

1.3 Wire Line Syntax

```
connect V1.p -> R1.1 -> C1.p -> VOUT.1
```

```
// Multiple wires from one node (junction):  
wire J1.1 -> R1.1  
wire J1.1 -> C1.p  
wire J1.1 -> LED1.A
```

1.4 Coordinate Rules

x increases left→right (0 to 14). y increases bottom→top (0 to 10). Space components at least 2 units apart. Sources on the left (x=0–2), loads on the right. GND below connected component, VCC above. Transistors/MOSFETs need 3 units of x-space for collector/emitter leads. Use fractional coords for sub-grid precision: at (2.5, 3.5).

2. Built-in Electronic Symbols

These symbols are always available — no import required.

Type ID	Category	Pins	Pin Names (DSL name)
VSource	Sources	2 pins	1:p 2:n
ISource	Sources	2 pins	1:p 2:n
Resistor	Passives	2 pins	1:1 2:2
Capacitor	Passives	2 pins	1:p 2:n
Inductor	Passives	2 pins	1:1 2:2
Diode	Semiconductors	2 pins	1:A 2:K
LED	Semiconductors	2 pins	1:A 2:K
Zener	Semiconductors	2 pins	1:A 2:K
Schottky	Semiconductors	2 pins	1:A 2:K
NPN	Semiconductors	3 pins	1:B 2:C 3:E
PNP	Semiconductors	3 pins	1:B 2:C 3:E
NMOS	Semiconductors	3 pins	1:G 2:D 3:S
PMOS	Semiconductors	3 pins	1:G 2:D 3:S
OpAmp	Semiconductors	5 pins	1:IN+ 2:IN- 3:OUT 4:V+ 5:V-
Ground	Connections	1 pins	1:1
VCC	Connections	1 pins	1:1
Node	Connections	1 pins	1:1
Switch	Misc	2 pins	1:1 2:2
Transformer	Misc	4 pins	1:P1 2:P2 3:S1 4:S2
JFET	Semiconductors	3 pins	1:G 2:D 3:S
Thyristor	Semiconductors	3 pins	1:A 2:K 3:G

2.1 Built-in Pin Quick Reference

```
// SOURCES (vertical – positive terminal top)
VSource(12V,DC) .p (pos/top) .n (neg/bottom)
ISource(1mA) .p (top) .n (bottom)

// PASSIVES (horizontal)
Resistor(10k) .1 (left) .2 (right)
Capacitor(100nF) .p (left+) .n (right-)
Inductor(10mH) .1 (left) .2 (right)

// DIODES (horizontal – current flows left→right)
Diode(1N4148) .A (anode/left) .K (cathode/right)
LED(red) .A .K
Zener(5.1V) .A .K
Schottky(1N5819) .A .K

// TRANSISTORS
NPN(2N2222) .B (base/left) .C (collector/upper-right) .E (emitter/lower-right)
PNP(2N3906) .B .C .E
NMOS(IRF540) .G (gate/left) .D (drain/upper-right) .S (source/lower-right)
PMOS(IRF9540) .G .D .S
JFET(J201) .G .D .S

// OP-AMP (5 pins)
```

```
OpAmp(LM741)      .IN+ (non-inv)  .IN- (inv)   .OUT   .V+ (pos supply)  .V- (neg supply)

// TRANSISTOR COMPOUND
Thyristor(TYN612) .A (anode/left) .K (cathode/right) .G (gate/bottom)
Transformer       .P1 .P2 (primary) .S1 .S2 (secondary)

// CONNECTIONS (single pin)
Ground            .1 (top – wire TO this)
VCC(5V)           .1 (bottom – wire TO this)
Node              .1 (junction/label point)
Switch            .1 (left)  .2 (right)
```

3. Building Electrical Symbols

These symbols are available after importing building_electrical_symbols.json via Library → Import JSON. All are tested and validated for ElectroDSL import.

3.x Data & AV

Type ID	Category	Pins	Pin Names (DSL name)
TVSocket	Data & AV	2 pins	1:SIG 2:GND
DataSocket	Data & AV	4 pins	1:TX+ 2:TX- 3:RX+ 4:OUT
TelephoneSocket	Data & AV	3 pins	1:A 2:B 3:OUT

3.x Distribution

Type ID	Category	Pins	Pin Names (DSL name)
ConsumerUnit	Distribution	7 pins	1:MAININ_L 2:MAININ_N 3:MAININ_E 4:OUT1_L 5:OUT2_L 6:OUT3_L 7:OUT4
MCB	Distribution	2 pins	1:IN 2:OUT
RCBO	Distribution	4 pins	1:IN_L 2:IN_N 3:OUT_L 4:OUT_N
RCD	Distribution	4 pins	1:IN_L 2:IN_N 3:OUT_L 4:OUT_N
IsolatorSwitch	Distribution	4 pins	1:IN_L 2:IN_N 3:OUT_L 4:OUT_N
FuseBoard	Distribution	4 pins	1:IN_L 2:IN_N 3:IN_E 4:OUT_L
ElecMeter	Distribution	6 pins	1:IN_L 2:IN_N 3:IN_E 4:OUT_L 5:OUT_N 6:OUT_E

3.x Fire & Safety

Type ID	Category	Pins	Pin Names (DSL name)
SmokeDetector	Fire & Safety	3 pins	1:L 2:N 3:INT
HeatDetector	Fire & Safety	3 pins	1:L 2:N 3:INT
FireAlarmBell	Fire & Safety	2 pins	1:L 2:N
FireCallPoint	Fire & Safety	3 pins	1:L 2:N 3:COM
CODetector	Fire & Safety	2 pins	1:L 2:N

3.x Heating & Controls

Type ID	Category	Pins	Pin Names (DSL name)
ThermostatRoom	Heating & Controls	5 pins	1:L 2:N 3:COM 4:HW 5:CH
BoilerController	Heating & Controls	4 pins	1:L 2:N 3:E 4:OUT

3.x Lighting

Type ID	Category	Pins	Pin Names (DSL name)
LightBulb	Lighting	3 pins	1:L 2:N 3:E
EmergencyLight	Lighting	3 pins	1:L 2:N 3:E
ExitSign	Lighting	3 pins	1:L 2:N 3:E
SpotlightFixed	Lighting	3 pins	1:L 2:N 3:E

3.x Security

Type ID	Category	Pins	Pin Names (DSL name)
AccessControlReader	Security	4 pins	1:V+ 2:GND 3:D0 4:OUT
CCTV	Security	3 pins	1:V+ 2:GND 3:VID

3.x Socket Outlets

Type ID	Category	Pins	Pin Names (DSL name)
SingleSocket	Socket Outlets	3 pins	1:L 2:N 3:E
DoubleSocket	Socket Outlets	3 pins	1:L 2:N 3:E
FusedSocket	Socket Outlets	3 pins	1:L 2:N 3:E
WatertightSocket	Socket Outlets	3 pins	1:L 2:N 3:E

3.x Specialist

Type ID	Category	Pins	Pin Names (DSL name)
EV_Charger	Specialist	5 pins	1:L1 2:L2 3:L3 4:N 5:E
SolarInverter	Specialist	6 pins	1:DC+ 2:DC- 3:PE 4:AC_L 5:AC_N 6:AC_E
BatteryStorage	Specialist	4 pins	1:DC+ 2:DC- 3:OUT+ 4:OUT-

3.x Switches

Type ID	Category	Pins	Pin Names (DSL name)
Switch1Way	Switches	2 pins	1:COM 2:L1
Switch2Way	Switches	3 pins	1:COM 2:L1 3:L2
SwitchIntermediate	Switches	4 pins	1:L1 2:L2 3:L3 4:L4
DimmerSwitch	Switches	2 pins	1:L 2:OUT
PIRSwitch	Switches	3 pins	1:L 2:N 3:SW
TimerSwitch	Switches	3 pins	1:L 2:N 3:OUT

3.x Wiring Accessories

Type ID	Category	Pins	Pin Names (DSL name)
DoorBell	Wiring Accessories	2 pins	1:A 2:B
CookerSwitch	Wiring Accessories	6 pins	1:IN_L 2:IN_N 3:IN_E 4:OUT_L 5:OUT_N 6:OUT_E
ShowerPull	Wiring Accessories	4 pins	1:IN_L 2:IN_N 3:OUT_L 4:OUT_N
ExtractorFan	Wiring Accessories	3 pins	1:L 2:N 3:E
ShedArmoured	Wiring Accessories	4 pins	1:L 2:N 3:E 4:OUT

4. Wiring Connection Rules

Always wire FROM a source pin TO a destination pin using the exact names in tables above.

4.1 Power Distribution Wiring

FROM component	Use pin	TO component	Use pin
ElecMeter	OUT_L / OUT_N	ConsumerUnit	MAININ_L / MAININ_N
ConsumerUnit	OUT1_L ... OUT4_L	MCB	IN
ConsumerUnit	OUT1_L ... OUT4_L	RCBO	IN_L
MCB	OUT	RCBO	IN_L (for downstream RCBO)
RCBO	OUT_L	load circuits	L (line feed)
IsolatorSwitch	IN_L / IN_N	IsolatorSwitch	OUT_L / OUT_N (downstream)

4.2 Lighting Circuit Wiring

FROM component	Use pin	TO component	Use pin
MCB	OUT	Switch1Way	COM
Switch1Way	L1	LightBulb	L
MCB	OUT	Switch2Way	COM
Switch2Way	L1	Switch2Way (2nd)	L2 (staircase circuit)
Switch2Way (2nd)	L1	LightBulb	L
PIRSwitch	SW	LightBulb	L
MCB	OUT	EmergencyLight	L (always live – no switch)
Ground	1	LightBulb	N (neutral return)
Ground	1	LightBulb	E (earth)

4.3 Socket Ring Main Wiring

FROM component	Use pin	TO component	Use pin
RCBO	OUT_L	DoubleSocket	L (ring start)
DoubleSocket	N	DoubleSocket (next)	L (ring continues)
DoubleSocket (last)	N	RCBO	OUT_N (ring return)
Ground	1	DoubleSocket	E (earth every socket)
FusedSocket	L	supply circuit	– (spur from ring)

4.4 Fire Alarm Wiring (24V DC loop)

FROM component	Use pin	TO component	Use pin
VSource (24V DC)	p	SmokeDetector	L (zone loop start)
SmokeDetector	N	SmokeDetector (next)	L (daisy-chain)
SmokeDetector (last)	N	Ground	1 (loop return)
VSource (24V DC)	p	FireAlarmBell	L
FireAlarmBell	N	Ground	1
FireCallPoint	L	supply	– (on detector loop)

4.5 Transistor/MOSFET Wiring

FROM component	Use pin	TO component	Use pin
----------------	---------	--------------	---------

Signal source	—	NPN	B (base via resistor)
VCC supply	1	NPN	C (via collector resistor)
NPN	E	Ground	1 (emitter to GND)
Signal source	—	NMOS	G (gate via Rg)
Load	—	NMOS	D (drain, load to VCC)
NMOS	S	Ground	1 (source to GND)
Input+	—	OpAmp	IN+
Input–	—	OpAmp	IN–
OpAmp	OUT	load / feedback	—
VCC	1	OpAmp	V+
Ground	1	OpAmp	V– (or neg supply)

5. Master AI Prompt Template

Copy and use this prompt with Claude, ChatGPT, or any LLM. Replace [DESCRIPTION] with what you want to draw.

You are ElectroDSL, an expert electrical schematic code generator.
Output ONLY valid ElectroDSL code. No markdown, no backticks, no explanation.
Start directly with CIRCUIT "Title" {

■■■ SYNTAX ■■■

```
CIRCUIT "Title" {  
    ID: TypeName(params) at (x, y) label="Name" [value="val"] [rotate=0|90|180|270]  
    wire FROM_ID.pinName -> TO_ID.pinName  
}
```

■■■ COORDINATE RULES ■■■

x: 0-14 (left→right) y: 0-10 (bottom→top)
Space components at least 2 units apart.
Sources/supply: left side (x=0-2). Loads: right side.
GND: 2 units below connected component. VCC: 2 units above.
Transistors/MOSFETs: allow x-spacing of 3 for collector/emitter pins.

■■■ BUILT-IN SYMBOLS & EXACT PIN NAMES ■■■

VSource(V,DC AC)	.p .n
ISource(A)	.p .n
Resistor(val)	.1 .2
Capacitor(val)	.p .n
Inductor(val)	.1 .2
Diode/LED/Zener/Schottky	.A .K
NPN/PNP	.B .C .E
NMOS/PMOS/JFET	.G .D .S
OpAmp	.IN+ .IN- .OUT .V+ .V-
Thyristor	.A .K .G
Transformer	.P1 .P2 .S1 .S2
Ground	.1 (wire TO this – GND reference)
VCC(V)	.1 (wire TO this – power rail)
Node	.1 (junction / net label)
Switch	.1 .2

■■■ BUILDING ELECTRICAL SYMBOLS (import building_electrical_symbols.json first) ■■■

SingleSocket/DoubleSocket/FusedSocket/WatertightSocket	.L .N .E
Switch1Way	.COM .L1
Switch2Way	.COM .L1 .L2
SwitchIntermediate	.L1 .L2 .L3 .L4
DimmerSwitch	.L .OUT
PIRSwitch	.L .N .SW
TimerSwitch	.L .N .OUT
LightBulb/EmergencyLight/ExitSign/SpotlightFixed	.L .N .E
ConsumerUnit	.MAININ_L .MAININ_N .MAININ_E .OUT1_L .OUT2_L .OUT3_L .OUT4_L
MCB	.IN .OUT
RCBO/RCD	.IN_L .IN_N .OUT_L .OUT_N
IsolatorSwitch	.IN_L .IN_N .OUT_L .OUT_N
FuseBoard	.IN_L .IN_N .IN_E .OUT_L
ElecMeter	.IN_L .IN_N .IN_E .OUT_L .OUT_N .OUT_E
SmokeDetector/HeatDetector	.L .N .INT
FireAlarmBell/CODetector	.L .N
FireCallPoint	.L .N .COM
TVSocket	.SIG .GND
DataSocket	.TX+ .TX- .RX+ .OUT
TelephoneSocket	.A .B .OUT
AccessControlReader	.V+ .GND .D0 .OUT
CCTV	.V+ .GND .VID
DoorBell	.A .B
CookerSwitch	.IN_L .IN_N .IN_E .OUT_L .OUT_N .OUT_E

ShowerPull	.IN_L .IN_N .OUT_L .OUT_N
ExtractorFan	.L .N .E
ShedArmoured	.L .N .E .OUT
EV_Charger	.L1 .L2 .L3 .N .E
SolarInverter	.DC+ .DC- .PE .AC_L .AC_N .AC_E
BatteryStorage	.DC+ .DC- .OUT+ .OUT-
ThermostatRoom	.L .N .COM .HW .CH
BoilerController	.L .N .E .OUT

■■■ WIRING RULES ■■■

1. Every Ground and VCC must be wired to at least one component.
2. Every power input pin (L, L1, IN_L, V+) must connect to a supply or upstream output.
3. Every component must have its earth (E) connected to Ground.
4. Ring mains: last socket .N returns to RCBO .OUT_N.
5. Fire alarm: daisy-chain detectors L→N→L→N→... return to GND.
6. Lighting: switch COM from MCB, switch L1 to lamp L.
7. Wire pin names are CASE-SENSITIVE: .IN+ not .in+, .OUT_L not .out_l.

Generate the DSL for: [DESCRIPTION]

6. Category-Specific Prompt Additions

Append one of these blocks to the master prompt for specialised circuits.

6a. Building Lighting Circuit

Additional: Generate a complete lighting circuit with:

- MCB (B6A) fed from CU.OUT1_L
- Switch1Way for each single-point light (MCB.OUT → SW.COM → SW.L1 → lamp.L)
- Switch2Way pair for 2-way switching (staircase)
- PIRSwitch for toilet/utility lights
- EmergencyLight always live (no switch, direct from MCB)
- ExitSign always live
- All lamp .N → GND, all lamp .E → GND
- GND nodes placed 2 units below each lamp cluster

6b. Ring Main Socket Circuit

Additional: Generate a ring main with:

- RCBO (B32A 30mA) from CU.OUT2_L
- Ring: RCBO.OUT_L → SK1.L → SK1.N → SK2.L → SK2.N → ... → RCBO.OUT_N
- Each socket: .L from previous, .N to next in ring, .E → GND
- FusedSocket for kitchen/appliance spurs (not on ring, separate spur)
- Label each socket with location (label="Kitchen", label="Desk 1")

6c. Power Electronics (transistors/MOSFETs)

Additional: Generate power electronics with:

- Gate resistor (Rg 100Ω) between control signal and MOSFET .G
- Freewheeling diode across load (Diode .K → VCC, .A → drain)
- Source resistor (Rs) from MOSFET .S to GND for current sensing
- Decoupling capacitor from VCC to GND near device
- All VCC rails: VCC component wired to supply voltage node
- All GND connections: Ground components wired to 0V node

6d. Fire Alarm System

Additional: Generate a fire alarm system with:

- MCB (B6A) for 24V PSU feed
- VSource(24V,DC) as the fire alarm PSU
- Zone 1 (office): SmokeDetector → SmokeDetector → HeatDetector daisy-chain
- Zone 2 (entrance): SmokeDetector → SmokeDetector → FireCallPoint daisy-chain
- Each zone: PSU.p → SD1.L → SD1.N → SD2.L → SD2.N → GND.1
- Sounders: PSU.p → BEL1.L → BEL1.N → BEL2.L → BEL2.N → GND.1
- Manual call points on zone loops
- Separate GND nodes for each zone

7. Complete DSL Examples

7a. Simple LED Circuit (built-in symbols)

```
CIRCUIT "LED Indicator Circuit" {
  V1:  VSource(5V,DC)    at (0, 4)  label="5V"
  R1:  Resistor(470)     at (3, 4)  label="R1"   value="470Ω"
  LED1: LED(red)         at (6, 4)  label="LED1" value="red"
  GND1: Ground           at (0, 2)
  GND2: Ground           at (8, 2)

  wire V1.p    -> R1.1
  wire R1.2    -> LED1.A
  wire LED1.K  -> GND2.1
  wire V1.n    -> GND1.1
}
```

7b. NPN Switch (built-in symbols)

```
CIRCUIT "NPN Transistor Switch" {
  VCC1: VCC(12V)         at (6, 9)  label="+12V"
  SIG:  VSource(5V,DC)   at (0, 4)  label="Ctrl"
  RB:   Resistor(10k)    at (3, 4)  label="RB"   value="10kΩ"
  RC:   Resistor(1k)     at (6, 7)  label="RC"   value="1kΩ"
  Q1:   NPN(2N2222)      at (6, 4)  label="Q1"   value="2N2222"
  LOAD: Resistor(100)    at (9, 4)  label="Load" value="100Ω"
  GND1: Ground           at (0, 2)
  GND2: Ground           at (6, 2)

  wire SIG.p  -> RB.1
  wire RB.2   -> Q1.B
  wire Q1.C   -> RC.1
  wire RC.2   -> VCC1.1
  wire Q1.C   -> LOAD.1
  wire LOAD.2 -> GND2.1
  wire Q1.E   -> GND2.1
  wire SIG.n  -> GND1.1
}
```

7c. 1-Way Light Switch Circuit (building symbols)

```
CIRCUIT "1-Way Lighting Circuit" {
  MCB1: MCB(B6A)         at (1, 6)  label="MCB-L1" value="B6A"
  SW1:  Switch1Way(6A)   at (4, 6)  label="SW-1"   value="1-Way"
  LP1:  LightBulb(LED)   at (7, 6)  label="LP-1"   value="9W LED"
  GND1: Ground           at (1, 4)
  GND2: Ground           at (7, 4)

  // MCB fed from consumer unit output (shown external)
  wire MCB1.OUT -> SW1.COM
  wire SW1.L1   -> LP1.L
  wire LP1.N    -> GND2.1
  wire LP1.E    -> GND2.1
  wire GND1.1   -> MCB1.OUT
}
```

7d. Ring Main Sockets Circuit (building symbols)

```
CIRCUIT "Ring Main – Office Sockets" {
  RCB01: RCB0(B32A,30mA) at (1, 5)  label="RCB0-R1" value="B32A 30mA"
  SK1:   DoubleSocket(13A) at (4, 5)  label="SK-1"   value="Reception"
  SK2:   DoubleSocket(13A) at (7, 5)  label="SK-2"   value="Desk 1"
  SK3:   DoubleSocket(13A) at (10,5)  label="SK-3"   value="Desk 2"
  GND1:  Ground           at (1, 3)

  // Ring: RCB0 → SK1 → SK2 → SK3 → back to RCB0
  wire RCB01.OUT_L -> SK1.L
  wire SK1.N       -> SK2.L
  wire SK2.N       -> SK3.L
  wire SK3.N       -> RCB01.OUT_N
}
```

```
// Earth every socket
wire GND1.1 -> SK1.E
wire GND1.1 -> SK2.E
wire GND1.1 -> SK3.E
}
```

8. Common AI Mistakes & Fixes

X Wrong pin name case	wire R1.Left -> C1.P	wire R1.1 -> C1.p (pin names are CASE-SENSITIVE)
X Wrong Ground wiring	wire GND1 -> C1.n	wire C1.n -> GND1.1 (use .1 on Ground)
X Missing earth connections	Socket with no .E wire	wire GND1.1 -> SK1.E (always earth sockets/lights)
X Ring main break	SK3.N not wired back	wire SK3.N -> RCB01.OUT_N (ring MUST close)
X OpAmp no supply	OpAmp without V+/V-	wire VCC1.1 -> U1.V+ wire GND1.1 -> U1.V-
X NPN wrong polarity	wire VCC -> Q1.E	VCC → collector (Q1.C), emitter (Q1.E) → GND
X Self-wiring	wire JP1.1 -> JP1.1	Never wire a component to itself – check IDs
X Duplicate component ID	R1: Resistor at (1,2)\nR1: Capacitor at (3,4)	Each ID must be unique
X Coords out of range	at (20, 15)	Keep x: 0-14 y: 0-10 for clean layout
X Missing CIRCUIT block	Starting with just wires	Must start: CIRCUIT "Title" { ... }
X Backtick wrapping	```\nCIRCUIT...\n```	Output raw DSL ONLY – no markdown fences

9. AI Prompt Checklist

- Tell AI: "Output ONLY raw DSL starting with CIRCUIT — no markdown, no backticks"
- Specify the circuit type clearly (lighting, ring main, fire alarm, power electronics...)
- Mention which symbol set: built-in only, or also building_electrical_symbols.json
- Confirm voltage/rating requirements (230V AC, 12V DC, 13A, B16A...)
- State number and location of each item (3 desk sockets, 2 lights with 2-way switching...)
- After generation: check every earth (.E) is wired to Ground
- After generation: check ring mains return to RCBO.OUT_N
- After generation: check no self-wires (wire X.pin -> X.pin)
- After generation: check no duplicate component IDs
- Import building_electrical_symbols.json before using building symbols
- Use Library → Import JSON to load custom symbols before pasting DSL

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